Yenah Bak

121 Pleasant Grove Rd, Ithaca, NY | 607-262-4434 | yenah990324@gmail.com

Education

Seoul National University (SNU) | Seoul, South Korea
Bachelor of Science & Bachelor of Arts [Biological Sciences / Psychology] ***Feb. 2024***

Publication

1. **Yenah Bak**, HK Kim, Kim, D. (In Preparation). "Negative salience regulates exploration through fastigial nucleus projecting dorsal raphe glutamatergic neurons."
2. **Yenah Bak**, HK Kim, JW Jeong, Kim, D. (In Preparation). [Review paper] "Systematic review and meta-analysis on Meta analysis: Emotions from physiology and neurological perspective"

Conference Presentations

1. **Yenah Bak,** S. Lee, J. Jeong, D. Kim (2026). *A brainstem-cerebellar circuit for environment-motor integration*
   Poster presentation at **FENS (2026),** Barcelona

2. **Yenah Bak,** S. Lee, J. Jeong, D. Kim (2024). *Serotonergic regulation of fastigial nucleus for muscle activity* regulation
   Poster presentation at the **Society for Neuroscience (2024)** Annual Meeting, Chicago, IL.

3. **Yenah Bak**, H Kim, Sungyon Kim. (2022). *Observation of Physiological Gastric Distension-activated Neural Population.*
   Oral and Poster presentation at the SNU Undergraduate Research Symposium.

Research Experiences

Korea Advanced Institute of Science & Technology (KAIST) | Daejeon, South Korea **Research Assistant**
(Advisor: Dr. Daesoo Kim) | *Apr 2024 – July 2026*
Primary Project: Functional dissociation of serotonergic regulation of motor
● Lead an independent project investigating the role of the serotonergic system in motor function regulation.
● Conducted fiber photometry, calcium and serotonergic biosensor recording, experiments to monitor serotonergic activity during stress and reward paradigms.
● Perform gait analysis and electromyography recording to identify a distinct subpopulation of dorsal raphe neurons that modulate motor functions.
● Manuscript in preparation (first author)

Collaboration Project (Lead: Dr. Kwanhoo Shin)*: Visual circuit principle of chase problem in hunting*
● Conducted stereotaxic surgery for optogenetic modulation during hunting trials
● Neural circuit tracing to validate topological connectivity
● Manuscript in preparation (middle author)

NeuroTobe Pharmaceuticals | Daejeon, South Korea **Researcher**
(Advisor: Dr. Sujin Chae) | *Jul 2023 – Mar 2024*
Project: Computational analysis of behavioral phenotypes for drug efficacy in transgenic mice
● High-dimensional data analysis to quantify drug effects on a transgenic mouse model: Integrated the **SUBTLE temporal analysis framework** to capture temporal components in animal behavior.
● Monitor drug effects on muscle activity with electromyography recording.

Seoul National University (SNU) | Seoul, South Korea **Undergraduate Researcher**
(Advisor: Dr. Alan Jung Park) | *Jan 2023 – Mar 2023*

●Lab Establishment: Assisted in the initial setup of laboratory infrastructure, including the procurement and calibration of stereotaxic, Deeplabcut analysis protocol, behavioral equipment.
●Protocol Development: Optimized standard protocol and procedures for rodent stereotaxic surgery and behavioral phenotyping.

Seoul National University (SNU) | Seoul, South Korea **Undergraduate Researcher**
(Advisor: Dr. Sung-yon Kim) | *Jul 2021 – Dec 2022*
Project: Mapping neural circuits of mechanosensory feedback in feeding behavior.
● Awarded the **SNU Undergraduate Research Grant** to lead an undergraduate project on stomach-distension feedback loops.
● Utilized **c-fos immunohistochemistry** to screen brain-wide neural activation following physiological stomach distension.
● Characterized specific inhibitory feeding circuits projecting from the **parabrachial nucleus (PBN)** to downstream targets, including the **periaqueductal gray (PAG)** and **paraventricular hypothalamus (PVH)**.
● Designed and optimized behavioral protocols to isolate physiological variables in mechanosensory feedback.

Computation-related Academic Experience

● **Language Model Implementation (C Language)** | *Applied Experimental Psychology (Fall 2022)*
● **Algorithm Design:** Programmed a multi-layer neural network from scratch in **C** to simulate grapheme-to-phoneme conversion (learning to pronounce words).
● **Project:** Implemented error-correction learning algorithms (backpropagation algorithm) to minimize output error.
● **Relevance:** Established the fundamental understanding of computational modeling and high-dimensional data processing that was subsequently applied to the **SUBTLE** analysis at NeuroTobe.
Computational & Data Analysis
● **Programming:** Python, MATLAB
● **Analysis:** Dimensionality Reduction (UMAP), SUBTLE framework
● **Tools:** ImageJ/Fiji, GraphPad Prism, Adobe Illustrator (for figure creation)

Honors & Awards

● SNU Undergraduate Research Grant (Project: Observation of Physiological Gastric Distension-activated Neural Population) | *2022*

● SNU Undergraduate Research Grant (Project: Comparative Analysis of UK/Korea Mental Health Systems) | *2019*

Skills & Techniques

Wet Lab & Physiology
● **In vivo physiology:** Fiber Photometry (calcium imaging), Stereotaxic Surgery (Optogenetics, chemogenetic, gastric balloon)
● **Histology:** Immunohistochemistry (IHC), c-Fos mapping, Perfusions, Vibratome/Cryotome, PRV circuit tracing
● **Behavioral Assays:** Gait analysis, Anxiety paradigms (Open Field, EPM), Behavioral test, Feeding assays.